Project Sun — Phase 2 Demo Reference

What the application really does, screen by screen, and the technology, legal, cost and design reasoning behind it. Written from the repository `project-sun-phase2-app-v2`, its local database and the build and audit chats (11 Sep 2026). UI labels are shown in **bold**; code names in `mono`. Figures marked “observed” come from real runs stored in `data/sun.db`.

Contents

1 · Product, users and value	1
2 · Journey map and data hand-offs	2
3 · Start: landing page and analysis form	2
4 · Understand: brand profile	2
5 · Landscape: topics and competitors	3
6 · Prompts: the 30-question instrument	4
7 · Observe: collecting ChatGPT and Gemini answers	4
8 · Measure: every section, label and metric	5
9 · How answers are analysed	6
10 · ChatGPT and Gemini: method and rationale	7
11 · Legal, terms and data handling	8
12 · Cost: Phase 2 and production	8
13 · Architecture, stack, storage, versioning	9
14 · Design decisions and build changes	9
15 · Limitations and honest answers	10
16 · Responsive and mobile behaviour	10
17 · Quick answers to likely questions	10

Demo run-sheet

1. **Before:** put `GEMINI_API_KEY` and `OPENAI_API_KEY` in `.env.local`. Check the OpenAI key with `npx tsx scripts/probe-openai.ts`. Start with `npm run build && npm start` (faster than dev) and open localhost:3000. Keep one fully collected project open as a fallback.
2. **Start:** paste a public site on the landing page and point out the live progress rail.
3. **Understand:** open a **Why?**. Edit one field (“saved as a new version”). **Confirm profile**.
4. **Landscape:** confirm 4–6 topics; reject a weak rival and add one you know; open **Why?**; **Confirm & continue**.
5. **Prompts:** show the three sections and a **Why?**. Try an edit that puts the brand into a Discovery prompt: it is refused, and the gate's reason (raw text) appears at the foot of the page. **Approve 30 / 30**.
6. **Measure: Collect automatically** (lanes, provisional banner, “so far”). By hand: **Copy prompt** → **Open ChatGPT** → paste answer and sources. Then switch **All / ChatGPT / Gemini**, open **?** and **See the answers**, and expand **How these results were measured**.
7. **If something stalls:** for a paused lane, ask by hand or show the pre-collected project. For a blocked crawl, edit the failed profile by hand.

The 45-second explanation

Project Sun measures how a brand shows up when buyers ask AI assistants. You enter a website. It crawls the public site safely and builds an **evidence-backed brand profile** you can edit and confirm. It suggests **topics and competitors**, which you confirm, and freezes that context. It then builds a validated set of **exactly 30 buyer questions**: 10 Discovery (no brand named), 10 Trade Offs (brand against one rival) and 10 Sentiment Analysis (brand by name). Those questions are put to **ChatGPT and Gemini**, automatically through the providers' APIs or asked by hand in the apps and pasted back. **Measure** then reports who gets recommended, head-to-head verdicts, how the brand is described and which sites are cited. Every figure shows its fraction, and every figure opens the exact answers behind it. Phase 2 covers measurement only. It has no recommendations or actions layer.

Key facts at a glance

Prompt set	30 = 10 Discovery / 10 Trade Offs / 10 Sentiment Analysis (`prompt-selection@3-10-10-10`)
Landscape gate	4–6 confirmed topics; ≥1 confirmed non-marketplace competitor
Crawl bounds	≤30 pages fetched, 26 used, depth 2, 90 s cap, robots.txt honoured
A run	30 prompts × 1 automatic route per assistant = 60 answers, 1 sample each (“Directional baseline”)
Observed timing	Crawl 9–38 s · competitor discovery 56–85 s · prompts ≈10–14 s (with Gemini) · Gemini lane 30 answers ≈2 min · ChatGPT on gpt-5-nano ≈2 min 11 s · first provisional figure ≈8–10 s
Observed cost	ChatGPT lane (gpt-5-nano) ≈ $0.0005–0.001 per answer (≈2–4¢ per 30); Gemini and everything else free tier or local
Engines	ChatGPT and Gemini only. Automatic = model APIs; by hand = consumer apps
Versions	`observation-analysis@9`, `measure-policy@1`, `competitor-discovery@4`, `intent-model@1`

1 · Product, users and value

Category: GEO / AEO (generative / answer-engine optimisation) measurement. Buyers increasingly ask assistants instead of searching. The question for a marketer is whether the answer names them, their rivals, or neither. **Phase 2 scope** is Understand → Landscape → Prompts → Measure. It does *not* include Diagnose, Actions or recommendations, content publishing, Perplexity or Google AI Overviews / AI Mode, billing, hosting, multi-user accounts, scheduled runs or repeated-sample statistics.

User	Question they bring	What they use / value they get
Marketing teams	“When buyers ask for options in our category, are we recommended, and who beats us?”	**Recommended unprompted**, **Share of voice**, **Who gets recommended**, **By topic**. A ChatGPT vs Gemini split. Each number opens the answers behind it.
Brand teams	“How do assistants describe us on returns, authenticity, sizing, support, value?”	**Reputation** section: positive, mixed, negative and neutral counts, a **By theme** tally and quoted answers. Factual answers are never counted as praise.
Agencies	“Can I onboard any client in minutes and defend the method?”	Works on arbitrary sites with no hardcoded brands. Evidence-cited profile. Frozen, versioned prompt sets. Published metric definitions and method notes to show clients.
GEO / AEO teams	“Is the prompt set realistic and balanced? What do assistants volunteer?”	30 validated prompts with a **Why?** panel per prompt: coverage, quality and selection reason. Also untracked brands that answers recommend, rank only when the answer is ordered, and the verdict per rival.
SEO teams	“Which sites shape the answers? Does our site get cited?”	**Sources**: most-cited domains, **Answers citing your site** and **Your share of cited links**. Sources are only present when answers carry them, which today means by-hand answers with pasted sources (see §10).
Enterprise marketing leads	“Can I trust this number? Is it comparable next quarter?”	An x-of-n fraction beside every percentage. **so far** labels until collection finishes. Missing answers are never counted as absence. The lineage line shows question version, analysis version and metric policy. The comparability rule is stated.

2 · Journey map and data hand-offs

The top navigation (`JourneyNav`) shows four stages, all clickable: **1 Understand · 2 Landscape · 3 Prompts · 4 Measure**. Collection used to be a separate "Collect" stage. It now lives inside Measure, and `/projects/:id/collect` redirects there. Long work runs in the background, and progress streams to the page over Server-Sent Events, with 2.5 s polling running alongside.

Stage / URL (under `/projects/:id`)	User does	System does	Stored	Passed to next stage
Start `/` → `/trial/new`	Enters website, **Market**, **Language**; clicks **Analyse my brand**	Normalises URL and blocks private addresses. Creates project. Starts crawl in background (HTTP 202). Redirects.	`projects` (url, registrable domain, market, language, capability-gate snapshot)	Project id; crawl already running
Understand `/understand`	Watches progress; opens **Why?**; edits fields; **Confirm profile**	Crawl → typed evidence → deterministic field resolvers → optional Gemini refinement → validation. Partial versions stream in.	`crawl_runs/pages`, `evidence_items/chunks`, raw page text files, `business_profile_versions`, `locations`	Confirmed profile version (brand, aliases, domains, offerings, audience, markets, archetype)
Landscape `/landscape`	Confirms, removes or adds 4–6 topics; confirms, rejects or adds competitors; **Confirm & continue to prompts**	Clusters topics. Discovers competitors from 4 sources, verifies their websites, scores and types them. Freezes the set.	`topics`, `entities`, `aliases`, `competitor_discovery_runs`, `competitive_set_versions` (snapshot + hash)	Frozen competitive-set version (topics + confirmed non-marketplace rivals)
Prompts `/prompts`	Reviews 30; **Why? / Edit / Regenerate / Replace / Deselect / Reject / Select**; **Approve**	Context snapshot → intent model → drafts → validators → dedupe → 100-pt score → select 10/10/10. Approval re-validates and freezes.	`prompt_context_snapshots`, `prompt_sets`, `prompt_candidates` (every candidate, state and reason)	Approved, frozen set of 30
Measure: collect `/measure`	**Collect automatically**, **Connect a key**, or **Ask ChatGPT & Gemini by hand**, then paste answers	Plans one prompt run per prompt per route. Background lanes per provider. Stores raw answer, then analyses it.	`collection_runs`, `prompt_runs`, `provider_responses`, raw answer files, `manual_capture_events`, `observation_analyses`	Versioned analyses (one per answer)
Measure: results (same page)	Switches **All / ChatGPT / Gemini**; opens **?** and **See the answers**	Metrics computed on every page load from analyses, planned runs and the frozen set. Deterministic; no model involved.	Nothing new: figures are recomputed, not cached	—

3 · Start: landing page and analysis form

Landing page (`/`): what each part says

- **Hero:** "Know where your brand stands in AI answers." A website field with **Analyze my brand**. The dashboard preview is labelled **Illustrative data**.
- **A new discovery layer:** search results versus a synthesised AI answer.
- **How Project Sun works:** 5 steps. The note "Phase 2 focus: Measure" says future Diagnose/Act features are not claimed.
- **Prompt intelligence** (neutral vs brand-aware prompts), **Competitive measurement**, **Evidence, not a black box** (metric → prompt → answer → brands → sources), **Industries**.
- **The measurement layer:** example cards (Presence, Share of Voice, Position, Citations, Coverage), marked "illustrative examples".
- **AI-answer surfaces:** ChatGPT and Gemini, "Automatic, or asked by hand". **Methodology**, 7-question **FAQ**, final call to action ("No card needed for this local Phase 2 trial").

Analysis form (`/trial/new`)

Field	Options	Where it goes
Website	Any public http(s) address. "example.com" gets https added.	Normalised. IP literals, localhost and `.local`/`.internal` rejected. Registrable domain extracted (`tldts`).
Market	Global, AE, US, UK, IN, SA, EU	Stored on the project only. **Not used downstream today**: markets come from site evidence and your edits (§15).
Language	English, Arabic, Hindi, French, Spanish	Passed to the crawl, used as the competitor-discovery locale, and recorded on every prompt. Generation instructions are English, so non-English wording has not been validated.

A URL typed on the landing page is carried over (`?url=`) and analysis starts automatically, guarded against double creation. Internally the app calls `POST /api/projects` then `POST /api/projects/:id/crawl` (202, fire-and-forget), then redirects to Understand. Errors are shown inline: "Public HTTP(S) URLs only — private and internal addresses are rejected."

4 · Understand: brand profile

What the user sees: "Your brand profile — Built from a safe crawl of your website. Review it, correct anything we got wrong, then confirm — everything downstream is measured against this." A progress rail ("Understanding {domain}", %) steps through **Website connected** → **Discovering site structure** ("N URLs found (M from sitemap)") → **Reading the most useful pages** ("x of y selected pages processed") → **Extracting evidence** ("N observations collected") → **Building the brand profile** (lists fields that have landed). The card header reads **Version N** ("still filling in…" while partial), then **Awaiting your confirmation / Confirmed · N pages analysed · N pieces of evidence · AI-assisted / evidence-only**. If the crawl is blocked, the page says "We could not analyse this website" and gives the reason: robots.txt disallows, private address, DNS failure or no pages fetched. It adds "You can still fill the profile in by hand below"; the failed version stays editable.

Field (UI label)	How it is generated	Editable as
Brand name + "Also known as"	Candidates ranked by evidence type: Organization schema 1.0 › WebSite schema .85 › og:site_name .8 › logo alt .6 › title segment repeated across pages .55 › page title .4 › domain .2 (last resort). Section names such as "HubSpot Blog" are folded into the brand.	Text input
Description	Without a key, one plain sentence assembled from evidence-backed fields ("Online retailer selling … for … in India."). With a Gemini key, 1–2 factual sentences written only from cited evidence, with hype words stripped. If too thin, it falls back to the assembled sentence.	Textarea (≤600 chars)
Industry	Closed taxonomy (NAICS-style sector › category). A Schema.org type pins it; otherwise weighted word-boundary keyword scoring. Below 25 confidence it shows "Not classified — please choose one".	Dropdown of taxonomy

Field (UI label)	How it is generated	Editable as
Business type	One of 12 archetypes: B2B SaaS, ecommerce/DTC, local service, dealership, restaurant, multi-location chain, healthcare, finance, marketplace, travel/hospitality, education, other. This drives prompt weights, what customers choose between, trade-off criteria, sentiment themes, geography emphasis, competitor weights and topic templates.	Dropdown
Brand identity	Curated trait vocabulary (Premium, Affordable, Sustainable, Enterprise-focused…) matched only in prose and headings. A trait found on one page is marked “check”.	Chips add/remove
Products & services	Category-level names ranked by cross-page support. Rejects SKUs, price filters (“Under ₹4999”), campaigns, “Products/Services”, and names from customer-logo pages.	Chips
Target market	Signals, strongest first: ccTLD › locale path (/in/) › hreflang set › store/location addresses › schema `areaServed` › currency › phone code › prose. Head-office address ≠ market; .com ≠ US; site language ≠ market. Hint shows “Scope: national…”.	Chips + level (city / region / country / multi-country / global)
Target audience	“For …” statements on pages where the business explains itself (not product listings).	Chips

Badges: Your edit (user source) · **High confidence** ≥75 · **Medium** ≥50 · **Low** >0 · **Needs review** 0. **Confidence is computed, never taken from a model:** 70 × strongest source type (JSON-LD 1.0 … body text lower), plus up to 18 for agreement across distinct pages, plus up to 8 for distinct source kinds, plus 6 when structured data and prose agree, minus 18 per conflict. **Why?** opens the evidence items: source type, page type, quoted text and URL. Also shown: **Regulated domain detected** (healthcare, finance or legal; advice-seeking prompts are then blocked) and **N notes from the analysis**.

Actions → internals: Edit profile is disabled while partial. **Save changes** creates a **new immutable version**: edited fields become source “user” at confidence 100 with evidence ids cleared, and edited paths are recorded so a re-analysis keeps them (`merge.ts`; no UI button for re-analysis). **Save & confirm** and **Confirm profile** mark the version confirmed (409 if still partial) and open Landscape.

Crawl internals: robots.txt parsed per RFC 9309 (user agent `ProjectSunBot/1.0`). Sitemap and index sampling. A planner scores URLs by page-type value with diversity quotas so product pages can't exhaust the budget. Same registrable domain only; 4 concurrent; 12 s timeout; 2 MB page cap. Playwright Chromium re-renders at most 3 JS-shell pages. JSON-LD, microdata, meta, headings and nav are extracted into ≤90 typed evidence items per page, plus 1,500-char text chunks.

Grounding guarantee: every substantive field must cite evidence ids that exist, and invalid ids are stripped. Gemini (`gemini-flash-lite-latest`, JSON schema, temperature 0.2) sees only a curated evidence pack with per-field budgets, labelled as untrusted website data. It may refine or reject the deterministic baseline but may not invent. Any failure falls back to “evidence-only”. **Passed on:** the confirmed version supplies topic seeds (offerings, customer jobs), the competitor search profile, the target entity (brand name and domains), and the markets used for prompt geography. The `locations` table is written from the crawl.

5 · Landscape: topics and competitors

Screen: “Topics & competitors — Confirm 4–6 topics and the competitors buyers genuinely compare. This context is frozen before prompts are created.” Discovery starts automatically on first visit, or again if the last run failed or found nothing. It is idempotent: a double call never duplicates topics.

Topics card

- Badge **N/4–6 confirmed**. Each topic has a name, a description (“Covers Running Shoes; Trail…”) and **Confirm** / **Remove**. **Add a topic…** creates it already confirmed (source user).
- **Generated from** profile offerings (which may name a topic) and customer jobs (which may only support one). Clustering compares the words that distinguish two names after removing a shared head noun (“Sales Software” vs “CRM Software”). At most 6. Below 4, archetype templates top up, described as “Archetype-standard topic… no direct on-site evidence found yet”.
- **Why editable:** topics come from the site’s own wording and can be product lines or merchandising (“Jordan”, “Air Max”, “Gift Cards”). Topics steer Discovery coverage and the **By topic** view in Measure.

Competitors card

- “Target: {brand}”; badge **Discovering and verifying** or **N eligible confirmed**. Each row: name, type (**Direct / Substitute / Marketplace / aggregator / Adjacent**), **N% match**, **Website verified**, reason text, source labels (**Website evidence · AI hypothesis · Open knowledge · Search discovery · Added manually**).
- **Why?** shows **Matched on** (dimensions), **Evidence** (verification summary) and **Confidence** High ≥75 / Medium ≥50 / Low. Stored confidence = 0.7 × score + 0.3 × source confidence.
- **Confirm**, **Reject** (**Remove** once confirmed). **Add a competitor…** plus relationship dropdown creates it **already confirmed** (confidence 100, “Added manually by operator”, type editable). With nothing found, the empty state shows “No verified suggestions yet” and **Retry automatic discovery**.
- Footnote: confirmed Direct, Substitute and Adjacent rivals are compared and tracked. Marketplaces “are where people buy rather than what they choose between, so they are never compared”.

Discovery pipeline (`competitor-discovery@4`)

1. **Sources in parallel**, each isolated so one failure can't stop the rest. (a) **Website explicit**: sentences on the site saying “vs / compared to / alternative to / switching from / instead of”; partner, integration and customer contexts are excluded; confidence 62. (b) **AI hypothesis**: Gemini JSON, Zod-checked, confidence capped at 80, up to 16. (c) **Open knowledge**, keyless: resolves the target via Wikidata official website (P856), drops its own sub-brands and siblings, and scores a category *signature* (IDF-weighted shared Wikipedia categories), sitelink prominence and CirrusSearch “morelike” text similarity. Throttled 190 ms / 1.1 s with a named user agent. (d) **Authorised search**: interface only, disabled. (e) **Observed AI answers**: type seam only.
2. **Resolve duplicates** by normalised name or domain; merge evidence and aliases.
3. **Verify website** where a domain is known: homepage plus up to 3 about/product pages, SSRF-safe, 7.5 s timeout. Result is verified, contradicted or failed. A failure is *not* a contradiction.
4. **Score**: structural affinity .34, offering .14, job/use-case .13, category .11, audience .07, geography .07, archetype .05, semantic .05, evidence .04. Archetypes override (local businesses weight geography .28–.30). Unobserved dimensions are dropped and weights renormalised, so a site that blocks crawlers isn't penalised.
5. **Type and reject**: direct if structural affinity ≥45 (or strong offering, job and category overlap); marketplace if directory or aggregator; not a competitor if supplier, partner or customer. Rejected: the target itself, its parent or its own product line; contradicted identity; a local business far from the market; weak evidence (total <28 and structural <40). The top 16 are stored; rejected ones are hidden.

Confirm & continue to prompts (server re-checks 4–6 topics and ≥1 non-marketplace rival; 400 otherwise) → `freezeCompetitiveSet`. It writes **CompetitiveSetVersion N** with a topic snapshot (name, brand-neutral concept, priority, evidence) and a competitor snapshot (every confirmed rival except marketplaces: name, domain, aliases, type, confidence, score breakdown, provenance), hashed with SHA-256. An unchanged hash reuses the previous version.

Why confirm or reject: model and knowledge-graph output is a hypothesis. The confirmed set becomes the **Share-of-voice**

denominator, the **Trade Off rivals** and the **Discovery forbidden-name list**, so the user must own it. Freezing stops later edits from silently changing prompts or past figures.

6 · Prompts: the 30-question instrument

Screen: "Prompt Intelligence — Review the balanced 30-question measurement instrument built from the frozen business, topic, competitor, audience, and market context." Badge **N / 30 selected**; counters Discovery / Trade Offs / Sentiment Analysis **N / 10**. While generating: "Building and validating the candidate pool — N candidates generated · N validated." Filters: search, topic, intent. Each row shows the text and badges (intent, geography, sentiment theme, **Quality N**).

Section (UI description)	Rules enforced in code	Example (Nike, approved set)
Discovery — "Brand-neutral buyer questions that test whether AI independently chooses the target."	Must not contain the brand, aliases, domain, plural, proprietary product lines, or any confirmed rival's name, short name or domain. Must invite a shortlist, not "what should I look for". At most one detail: none, a situation, an audience written in first person, a budget, a quality, or a city for businesses people visit. Asks for what this business type's customers choose between (a product, a restaurant, a clinic…).	"Which running shoes are worth buying under ₹15,000?" · "Can you suggest some good football boots?"
Trade Offs — "Buying-decision comparisons against confirmed competitors (never marketplaces)."	Names the brand and exactly one frozen rival, using the name people say. Rivals come from the top 5 by confidence, with +8 for direct. The prompt must carry a decision factor. The model first declares which needs each rival competes on, and pairs outside those are discarded in code. Location-dependent comparisons ("closer to my office") are rejected.	"Nike or Adidas for a gym t-shirt that stays dry during heavy lifts?"
Sentiment Analysis — "Neutral branded questions about business-appropriate reputation themes."	Names the brand only. Tied to a customer concern or theme (returns, authenticity, sizing, support, delivery…). No leading or presupposing phrasing.	"Is it safe to buy Nike products online without getting knockoffs?"

Action	What happens internally
Why?	**Context** (coverage keys: category, detail, rival, concern, register), **Brand policy** ("Target and confirmed competitors forbidden" / "Target required"), **Scoring** ("Quality x/100 · Coverage y/100 · Diversity z/100"), **Selected because**.
Edit → Save and revalidate	Re-runs the same hard gates: leakage, rival named, factor or theme present, topic in frozen set, geography in scope, near-me, leading, multi-question, relative dates, regulated advice. Also checks for an exact duplicate and re-scores. The original text is kept; the edit is stored as `editedText`. A failure shows the reason (400).
Regenerate	Prompt written by the model: Gemini writes 3 new wordings that keep brand, rival and the single detail (temperature 0.9). Template prompt: re-realised from the same frame. New wordings pass validators, near-duplicate (≥0.72) and scoring, then are added to **alternatives**. They are not auto-selected. Without Gemini: "New wordings could not be written right now…" (409).
Replace	Picks the best validated alternative in the section (same rival +3, theme +2, category +2, topic +1, then score). The current prompt becomes REJECTED and the alternative SELECTED. Nothing changes if there is no alternative.
Deselect / **Reject** / **Select**	Deselect returns it to alternatives, so the section drops to 9. Reject marks it REJECTED with a reason. Select from **Review N validated alternatives** is refused if the section already has 10.
Approve 30 / 30 prompts	Server re-checks exactly 10/10/10, natural-language and policy validity of every final text including edits, and no exact duplicates. Then: USER_APPROVED, set status approved, `frozenHash`, approvedBy, approvedAt. The set can no longer change (409). Shows "Prompt set approved and frozen" and **Measure these 30 prompts** →. Approval does not start collection. A whole new set is API-only: `POST /prompts/generate?force=1` creates version N+1 and leaves old versions untouched.

How prompts are generated (lifecycle GENERATING → CANDIDATE → VALIDATED → SHORTLISTED → SELECTED → USER_APPROVED, or REJECTED with reason)

1. **Frozen context snapshot** (profile + competitive set, SHA-256), so regeneration can't pick up changed context. **Coverage policy** `prompt-selection@3-10-10-10`, with per-archetype intent weights, geography emphasis, trade-off criteria and sentiment themes.
2. **Customer intent model** (1 Gemini call, temperature 0.4): 6–9 shopper-level needs, each with situations, qualities, audiences and a budget in local currency. Needs are grounded in the site's navigation vocabulary; ungrounded needs are dropped. Plus 5–8 purchase concerns and 5–7 decision factors. Brand and rival names are banned.
3. **Three section calls in parallel** (Discovery temperature 0.8 with code-planned details; Trade Offs 0.6 with rival–need pairs; Sentiment 0.8), ≈26–32 drafts each, one retry if a section comes back empty. The **deterministic floor** always runs: frame planner and realiser with typed concepts, so "a reliable running" can't be built. Without a key, the floor is the only source.
4. **Validate** (policy + natural language: not analyst voice, no internal labels, no dangling slots, ≤34 words) → **dedupe** (exact, plus lexical similarity ≥0.72 within the same context) → **score out of 100**: semantic coherence 25, natural buyer language 20, business relevance 15, intent clarity 10, measurement usefulness 10, context appropriateness 10, specificity 5, evidence 5, minus redundancy up to 25. Capped at 25 if incoherent, 45 if unnatural; minimum 55 to be selectable.
5. **Select greedily per section**: 0.55 quality + 0.30 new coverage + 0.15 diversity − repetition penalty. Caps: about slots ÷ rivals + 1 per rival, one register (question / request / keyword) ≤60%, category share. Model-written drafts fill first; templates only fill gaps. Up to 3 regeneration retries if a section is short.

Observed: 275–320 candidates, 65–80 duplicates removed, 4 Gemini calls, ≈10–14 s. Benchmark with no key: 13 archetypes, 30 each, zero brand leakage, zero duplicate text, 0.2–1.2 s. **Location effect:** a place appears in prompt text only if it is a confirmed location at the right level. City-level Discovery is used only for businesses people visit. "Near me" is rejected unless geography is controlled. The budget currency follows the primary market.

7 · Observe: collecting ChatGPT and Gemini answers

Before any run exists, Measure shows "**Collect the answers** — Ask ChatGPT and Gemini the approved questions. Results appear here as soon as the first answers arrive." Once a run exists, the **observation console** sits in a collapsible Collection area at the top of Measure. It opens by itself when nothing has been collected or a run is active. The header button reads **Hide collection / Continue collecting / Collect again**. The console heading says "Collect answers" or, for example, "X of 60 collected automatically · Y of 60 asked by hand".

Console element	What it does (and what happens internally)
Collect automatically / **Collect N remaining**	Disabled without a key. It creates the run if there is none, choosing one automatic route per assistant. It adds automatic routes to a by-hand-only run. It moves unfinished ChatGPT questions from Groq or Cerebras to the preferred route. Then it starts the background runner (`POST /collection-runs/:id/collect`, 202).
Connect a key → "Turn on automatic collection"	Provider: **Gemini (Google AI Studio key)**, **ChatGPT (OpenAI key)** or **ChatGPT (Cerebras key)**, plus a password field. Keys are saved in local SQLite `provider_credentials`. Environment variables in `.env.local` win. The status API returns true/false only, never the key. Takes effect without a restart.
Ask ChatGPT & Gemini by hand	Plans 30 × {`chatgpt_manual`, `gemini_manual`} in the **same run**, already waiting for import. Automatic and by-hand evidence share one measurement.
Lane cards (one per route)	Show name, **automatic** or **by hand** chip, done/total and a status line: **Asking now** · "Next question in 12s, pacing to the free limit" · "Paused: today's free limit is used up. Collects again after 7:47." · "Paused: OpenAI is limiting requests for a moment." · **Done** · "N waiting for you" / **All captured**. Also "about N min left" and "N not collected yet: left out of the results, never counted as a miss".
Latest answers	The 6 newest answers: question, engine ("by hand"), time ago and a 2-line excerpt. Markdown is stripped for display only; the stored answer is untouched.
Ask by hand (tabs ChatGPT / Gemini)	Guidance: "Use a temporary or logged-out chat so your own history does not shape the answer." Four questions at a time (**Show all N questions**). Each row has a section label, **Copy prompt** (clipboard only), **Open ChatGPT** ↗ / **Open Gemini** ↗ (a new tab to chatgpt.com or gemini.google.com/app) and **Paste answer**.
Paste form	**Full answer** (≥20 chars). **Sources shown — one link per line** (counts recognised links). **Sources**: I copied every source shown / some of them / No sources were shown / I didn't copy sources. **Did it search the web?** Yes / No search shown / Not sure. **Model shown — optional**. Checkbox "The answer was cut off, or I only copied part of it". **Save observation**. Choosing "every source" with no links is refused. These assertions decide which denominators the answer may enter.
Footnote	"Automatic or by hand? Automatic answers come straight from OpenAI's and Google's models rather than the ChatGPT and Gemini apps, so they can differ from what a person sees…"

Planning and background runner

- A run needs the latest prompt set **approved with exactly 30**. Route preference, one per assistant: Gemini → `gemini_free_api`; ChatGPT → `openai_api` › `cerebras_free_api` › `groq_free_api`. With no key, the API returns 409 `no_collection_route`.
- One `prompt_run` per prompt × route. Idempotency key = SHA-256(project | set | prompt | route | sample 1), so adding routes never duplicates.
- The runner lives inside the Next.js server process. Lanes run concurrently, one per provider. Queue order is **Discovery** → **Trade Offs** → **Sentiment**, so headline numbers settle first. Progress events stream at most every 750 ms.
- Each automatic call sends a fixed system line ("buyer-style question… answer naturally… do not mention that this is a test") plus the prompt (edited text if edited). On success: the raw text is written, the `provider_responses` row is saved, then the answer is analysed and the run marked `succeeded`. An analysis error never loses the answer.
- State machine: `planned → claimed → running | awaiting_import → succeeded`. Failure states `rate_limited / failed / incomplete` return to `planned` on retry. `succeeded` and `policy_disabled` are terminal. Full attempt history is kept.

Lanes and pacing

Route	Parallel	Pacing	Observed
`gemini_free_api`	3	≤14 req/min window; honours stated *retryDelay*; 503 back-off 1.5/4/9 s	30 answers ≈2 min
`openai_api`	4	≤60 req/min (tier 1 allows 500)	30 ≈2 min 11 s; ≈17 s/answer
`cerebras_free_api`	1	≤5 req/min (1M tokens/day)	Account returned 402
`groq_free_api`	1	Token budget: waits for the 8,000 tokens/min budget reported in headers (assumes 2,600/answer)	30 ≈6.4 min; daily quota ran out once

Retries: up to 4–5 for rate limits and 2 for transient errors. A lane **pauses** when the stated retry-after exceeds 3 min, or after 4 refusals in a row. Its rows stay uncollected and a later Collect resumes them.

Failures and missing observations

Situation	Handling
Rate limit / quota	Retry after stated delay → pause lane; uncollected rows are **excluded**, never zero
Timeout, network, 503	Retried (transient)
No key, auth failed, model gone, `billing_required`, HTTP 400–404	Permanent `failed`; no retry loop
Server restarted mid-call	Orphaned `running` rows marked failed ("interrupted") and re-queued on next Collect
Answer hit token cap	Stored as `truncated` and excluded from presence (a brand after the cut is unobserved)
Switch provider mid-run	Unfinished rows retired as `policy_disabled` and re-planned on the new route; collected answers kept; planned stays 30

What every answer records (`provider_responses`)

Route (`api` / `consumer_app`) and method (`automated` / `manual`). Model shown (e.g. `gemini-flash-lite-latest`, `gpt-5-nano-2025-08-07`, `openai/gpt-oss-120b`). Search state (on / off / unknown / not_applicable). Source URLs plus source capture state (complete / partial / none_present / not_captured / unknown). Answer capture (complete / truncated). Locale, latency, observed time, adapter version, `route-policy@1`, and SHA-256 of the raw text. A by-hand paste also writes a `manual_capture_events` audit row (operator, checklist).

Where prompts and answers can be seen: Prompts page (all 30 plus alternatives) · console **Latest answers** and by-hand queue · any Measure figure → **See the answers** drawer (question, engine, section, full answer, sources; first 25) · **How these results were measured** · `GET /api/provider-responses/:id`.

8 · Measure: every section, label and metric

Page header and controls

UI element	Meaning
"AI visibility — How ChatGPT and Gemini talk about {brand}"	The brand is the target entity. Everything below uses only that run's answers.
Collecting / Partial results / Complete	Runner active / some planned answers missing / all planned answers in

Headline cards

Card	What it says
Recommended unprompted	"Named in x of n answers to questions that never mentioned {brand}" = Discovery visibility
Share of voice	"{brand} took x of n brand mentions in those answers"

UI element	Meaning
"X of N answers collected · Y asked by hand"	Coverage. N counts automatic routes when present; by-hand answers are extra and don't make a finished run look unfinished.
Directional baseline	One sample per prompt per route (`SAMPLING_POSTURE`)
All · ChatGPT · Gemini (with counts)	Recomputes every figure on that assistant's answers. **All** pools answers with equal weight; it is not an average of the two. By-hand answers count toward their assistant.
Sticky banner "Collecting: results update as each answer arrives" / "Partial results…"	Progress bar, ETA, paused lanes. "Figures marked 'so far' use only the answers collected. Missing answers are left out of every figure, never counted as the brand being absent."

Display rules

- Denominator <5: headline shows **"x of n"** + "a small sample" instead of a percentage.
- Zero denominator → "—" / "No answers to count yet". That is N/A, **never 0%**.
- While provisional: **so far** tag and dashed card borders; each section shows "N answers counted so far".
- **?** opens the published definition (**What is counted · Out of · Which questions · Missing answers**), served with the numbers from `measure/policy.ts`.
- **See the answers** → / any bar or row opens the evidence drawer: exact question, engine ("ChatGPT, asked by hand"), section, full answer and sources (first 25).

Card	What it says
Head-to-head "W–L"	"{brand} preferred in W, a rival in L, K with no clear winner" (Trade Offs)
Reputation "N positive"	"n negative, n mixed, n factual or neutral" (Sentiment Analysis)

Sections

Section / panel	What it represents
Discovery "When a buyer doesn't name {brand}" → **Who gets recommended**	Bars: answers naming each tracked brand ("An answer naming a brand five times counts once"), **you** chip, share %. **Not named yet**: tracked rivals with 0. **Also named, but not tracked**: untracked brands in ≥2 answers, not in share of voice ("Add a brand to your competitors to track it in the next run").
More from these answers	**{brand} named first**; **Answers naming no brand at all**
By topic	"named in x of n" per confirmed topic + **Most named** (counts, because each topic has few questions)
Trade Offs "When a buyer compares {brand} with a rival" ("only the verdict counts here, not the mention")	**Verdicts**: {brand} preferred / No clear winner / Verdict unclear / Rival preferred / Didn't compare them. Mini-stats **Gave a verdict** and **{brand} won, when decided**. **By rival**: won · lost · open.
Reputation "When a buyer asks about {brand} by name"	**How the answers describe it**: Positive / Mixed / Negative / Neutral / **Factual, no opinion** ("never counted as positive"). **By theme**: +/–/± per aspect actually discussed.
Sources "Which sites shape the answers"	If none: "None of these answers listed the pages they used…". Otherwise **Your own site** (three citation metrics) and **Most-cited sites** (**yours**/**rival** tags; answers · links).
How these results were measured	Per-route method sentence with collected counts (e.g. "ChatGPT, asked automatically: … gpt-5-nano … without web search"), app-vs-API caveat, sampling caveat, pooling rule, comparability rule, lineage "questions vN · observation-analysis@9 · measure-policy@1".

Metric definitions (`measure-policy@1`). Only `ELIGIBLE` answers count; missing, uncertain or truncated answers are excluded from both numerator and denominator.

UI label · key	Counts (numerator)	Out of (denominator)	Questions
Recommended unprompted · `discovery_visibility`	Answers where the brand is present (exact or strong match), once per answer	All eligible Discovery answers, including brand-free ones	Discovery
Share of voice · `discovery_share_of_voice`	Brand presence events (one per answer × brand)	Presence events of brand + every frozen rival (untracked brands excluded)	Discovery
{brand} named first · `first_choice_rate`	Answers where the brand holds explicit rank 1	Rank-eligible answers only (numbered list or ordinal prose)	Discovery
Answers naming no brand at all · `answer_brand_free_rate`	Answers naming no tracked brand	Eligible Discovery answers	Discovery
Gave a verdict · `tradeoff_decisiveness`	Answers preferring one side	Trade Off answers containing a comparison (ties/unclear stay in)	Trade Offs
{brand} won, when decided · `tradeoff_target_win_rate`	Answers preferring the brand	Decisive answers only	Trade Offs
Answers citing your site · `owned_citation_rate`	Answers citing ≥1 URL on the brand's domain	Answers with trustworthy source capture: "all sources copied" or "none shown"	All sections
…of answers that showed sources · `owned_citation_rate_when_sourced`	Same	Answers that displayed ≥1 source	All sections
Your share of cited links · `citation_share`	Cited URLs owned by the brand (by URL volume)	Cited URLs owned by brand or frozen rivals	All sections

9 · How answers are analysed (deterministic, no LLM judge)

1. **Segment** into sentences and clauses with offsets preserved. Negation scope follows NegEx: cues such as "wouldn't, isn't, avoid" are cut off at "but, however, while". Hedges ("may, arguably") are flagged. Numbered/bulleted list and "in no particular order" are detected.
2. **Mentions**: link only the target and **confirmed** entities. Surfaces are canonical name, aliases, short names the domain supports ("Brooks" for Brooks Sports, capitalised only) and plurals ("Nikes"). Longest match wins; ≥3 chars. A match followed by a corporate word ("Puma Energy") is weak, and weak matches never count as presence.
3. **Role** per mention from its clause: RECOMMENDED (recommend verb, list framed as options, "popular brands include…"), DISCOURAGED, COMPARED, BACKGROUND (founded, revenue…), MENTION_ONLY or UNCLEAR.
5. **Rank** only if explicitly ordered (numbered list, "top pick… then…"). An unordered list or prose is never ranked.
6. **Sentiment** per brand per clause, as aspect / category / opinion / polarity. Categories: quality, price/value, comfort/fit, performance, design, delivery, returns, support, reliability, ease of use, selection, authenticity, project topics, overall. Negation flips polarity; praise plus criticism = Mixed; facts only = no sentiment; clauses comparing two brands are skipped.
7. **Citations** → registrable domain → TARGET_OWNED / COMPETITOR_OWNED / THIRD_PARTY. Kept separate from mentions: being cited isn't being named.
8. **Untracked brands**: capitalised names written mid-sentence in an option-like role and never lower-case elsewhere in the answer. Never counted in denominators.

4. **Comparisons** as (subject, object, aspect, opinion, preference). A selection verb (“I’d pick”) picks the entity after it; the verdict can come in the next sentence. “If X…, if Y…” is a tie. “A isn’t as good as B” favours B. A bare comparative is weak. Brand-relative outcomes: preferred / rival / tie (mixed aspects) / unclear / no comparison.

9. **Validate** that every span matches the raw text. **Eligibility**: `NEEDS_REVIEW` if a span is broken, a rank is impossible, a brand match is uncertain or the answer is truncated; otherwise `ELIGIBLE`. A run with no answer is `NOT_CAPTURED`. Stored as a new `observation_analyses` row (`observation-analysis@9`). The raw answer is never modified, and the current version wins.

Why deterministic: reproducible numbers, auditable spans and no per-answer model cost. Re-analysis (`scripts/reanalyse.ts`) re-derives figures without re-collecting. Real-output fixes during the build moved the same 60 Nike answers from 51 to 60 analysable, and share of voice from 39% to 25%.

10 · ChatGPT and Gemini: method and rationale

Short answer: two ways to collect, both feeding one run, and every answer is labelled with how it was collected. (1) **Automatic**: the providers’ **model APIs** (developer endpoints, not the apps). (2) **By hand**: a person asks in the **ChatGPT or Gemini consumer app** and pastes the answer and sources back. The app never automates, scrapes or logs into chatgpt.com or gemini.google.com.

Route (code)	Shown in UI as	How	Model	Search / sources	Limits / status
`chatgpt_manual`	ChatGPT, asked by hand	Operator asks at chatgpt.com (temporary chat advised), pastes answer and sources, states capture quality	Whatever the app uses (optional field)	As the app shows; operator states whether it searched	Always available, no key. Slow; depends on the operator.
`gemini_manual`	Gemini, asked by hand	Same, at gemini.google.com/app	As app	As app	Always available
`openai_api`	ChatGPT, asked automatically	OpenAI Chat Completions (plain `fetch`). Key in `.env.local` or UI. **Preferred ChatGPT route.**	`gpt-5-nano`; dated snapshot recorded (e.g. `-2025-08-07`)	No web search → no sources	Paid, ~0.1¢/answer; 6,000 max output tokens (includes hidden reasoning); 4 parallel
`cerebras_free_api`	ChatGPT, asked automatically	Cerebras free tier (OpenAI-compatible)	`gpt-oss-120b` (OpenAI open-weight)	None	5 req/min, 1M tokens/day; this account returned 402 (payment required)
`groq_free_api`	ChatGPT, asked automatically	Groq free tier	`openai/gpt-oss-120b`	None	8K tokens/min, ~200K/day (≈80–120 answers/day)
`gemini_free_api`	Gemini, asked automatically	Google Gemini API via `@google/genai`, AI Studio key	`gemini-flash-lite-latest` (alias; resolved name stored)	Tries Google Search grounding first and falls back to plain if the key refuses. All 298 stored answers are ungrounded with no sources.	Free tier. Flash-Lite chosen because the full Flash free daily quota was exhausted by a single 30-prompt run.
`consumer_browser_automation`	—	Policy `enabled=false`: provider terms prohibit automated extraction, and both apps disallow their chat paths in robots.txt.			
`licensed_partner_collector`	—	Future production route with contractual rights. Seam only; not implemented.			

What was considered, and why not (yet)

- **Browser automation** of the consumer apps (Playwright, extension): the most faithful answers, but blocked by terms and robots rules, plus account, CAPTCHA and UI-change fragility. Rejected.
- **Third-party collection vendors** (scraping proxies): they don't cure the provider's terms and would need written rights, provenance and deletion terms. Rejected for Phase 2.
- **OpenAI API with web search** (Responses API): closer to ChatGPT search and returns sources. It costs ~$10 per 1,000 searches plus page tokens. Not enabled; a straightforward next step.
- **Gemini API with Search grounding**: returns citations, but Google's Gemini API Additional Terms restrict analysing grounded results, and it is unavailable on the free tier.
- **Free GPT routes**: GitHub Models' free GPT-4o was retired 30 Jul 2026 (returns 410). Groq and Cerebras serve only OpenAI's open-weight model with small daily caps, which is why the ChatGPT lane moved to paid `gpt-5-nano` for a reliable demo.
- **ChatGPT Plus / Google AI Pro subscriptions**: consumer app access only, with no API credit or automation right. Used only for asking by hand.
- **How competitors describe it** (their public statements, not verified): Peec documents simulated browser sessions; Profound says it monitors consumer engines directly and uses licensed opt-in panels; Scrunch mixes browser automation and official APIs.

Why this approach

- **Lawful and honest:** no automation of consumer surfaces, and API answers are never presented as app answers (route and method are stored on every row).
- **Demo-reliable and near-free:** automatic lanes give 60 answers in about 2 minutes. By-hand answers add app-true evidence to the same run.
- **User-facing naming:** marketing users see only **ChatGPT** and **Gemini**, plus “automatic” or “by hand”. Model and provider details (gpt-5-nano, gpt-oss on Groq) sit only in **How these results were measured**.

Current limitations

- API ≠ app: no personalisation, memory, app system prompt or app model choice. `gpt-5-nano` is a small model, not what ChatGPT Plus users get.
- No web search on automatic lanes → **no sources** → the Sources section stays empty unless by-hand sources are pasted.
- One sample per prompt. No control of the location an answer is asked from, and it isn't recorded in the UI. Model aliases can drift, though the resolved name is stored.
- By hand is slow (plan estimate 45–120+ min for 60 answers), relies on the operator's honesty, and a saved answer can't be corrected in the UI.

What production would change

Written provider or licensed-partner rights for app-faithful collection, or clearly labelled API-with-search lanes. 3–5+ samples per prompt, scheduled. Location and language controlled execution. Recorded session and persona profiles. More engines (AI Overviews, AI Mode, Perplexity,

Copilot, Claude) only where rights exist. Operator tooling with QA double-capture. A durable queue with workers. Tracking of model version drift.

11 · Legal, terms and data handling (design reasoning, not legal advice)

Topic	Position taken and why	What is implemented
Automated browser access / scraping of ChatGPT or Gemini	OpenAI's consumer Terms of Use (effective 1 Jan 2026) prohibit users to "automatically or programmatically extract data or Output". Google's Terms restrict automated access that goes against machine-readable instructions, and both apps disallow their chat paths in robots.txt. A subscription grants no automation right, and a vendor doing it for you does not cure the terms.	No DOM reading, cookies, injected scripts, simulated clicks or submission. `consumer_browser_automation=false`. The console only copies text and opens a tab.
User-mediated / manual collection	A human asks in their own session and copies what they see: defensible for a research or demo baseline when provenance is recorded. It is **not** a blanket commercial right; some providers, e.g. Perplexity, restrict even manual monitoring for commercial purposes. Counsel should review before commercial scale.	By-hand route with operator assertions and a `manual_capture_events` audit row. Temporary chat advised to avoid personal history.
APIs vs consumer responses	APIs are the supported programmatic path under each provider's API terms, but they are a different experience, so they are labelled and never relabelled as app answers. OpenAI API data is not used for training by default; abuse logs are kept ~30 days. Gemini free-tier content may be used to improve Google products, so only public website text and synthetic buyer prompts are sent.	Route, method, model, adapter version and policy version stored per answer. Official terms URLs recorded in `route_policies`.
Search grounding	Gemini API Additional Terms (effective 23 Mar 2026) restrict analysing grounded results and collecting their links for other purposes; Google retains grounding data ~30 days.	**Gap:** the Gemini lane *requests* grounding whenever the key allows. Free keys refuse it, so 0 grounded answers are stored. A capability flag `geminiGroundedSearchAnalytics=false` exists but the adapter doesn't enforce it. Gate this before using a billed key.
Website crawling	Only the submitted site's public pages, respecting its robots rules; transparent, bounded fetching.	RFC 9309 robots (crawl-delay capped), UA `ProjectSunBot/1.0`, same registrable domain, ≤30 pages, SSRF guard, Playwright only for that site. Competitor verification fetches ≤4 public pages per candidate **without a robots check** (gap). Wikipedia/Wikidata APIs with UA and throttling.
Prompt injection	Website text is untrusted data, never instructions.	System prompts say so. The model has no tools or secrets, and its output is schema-checked and evidence-validated.
Raw response storage	Raw text is kept so every figure is auditable and re-analysable; redisplaying provider output in a hosted product needs a rights review.	Verbatim text in SQLite and in `evidence-store/<project>/<sha256>.txt`; shown in drawers. No screenshots. No personal data by design (synthetic prompts); operator recorded as `interview_operator`.
Data retention	Local-only Phase 2; retention should be enforceable per route or customer in production. Provider-side retention (OpenAI API ~30 days; Gemini Apps activity settings) is outside the app's control.	`SUN_EVIDENCE_TTL_DAYS=30` and purge/export/delete functions exist, but **nothing calls them** (no scheduler or UI). Route policies list 30-day retention. Keys pasted in the UI sit in plaintext in local SQLite (env vars preferred). Data folders are git-ignored.

Suggested wording: "We only automate through provider APIs, and we label those answers as API answers. The consumer apps are observed by a person, never by a bot, because their terms forbid automated extraction. For production we'd need written collection rights or licensed partners, and legal review of storage and redisplay."

12 · Cost: Phase 2 and production

Phase 2 is effectively free except the ChatGPT API lane. It runs on a laptop with SQLite and local files; there is no cloud, queue or database server. Gemini is used on the free tier. Analysis and measurement are local code with no per-answer model cost. Asking by hand uses subscriptions the operator already has. Prices below are OpenAI list prices recorded in the build (`gpt-5-nano` $0.05 per 1M input tokens, $0.40 per 1M output tokens) and are **estimates**.

Stage	What it calls or uses	Phase 2 cost	Production cost driver
Website analysis	≤30 HTTP fetches; headless Chromium for ≤3 JS-shell pages; local CPU; 0–1 Gemini call for profile synthesis	$0 (free tier)	Crawl compute, bandwidth, headless browser capacity; LLM tokens per project (small)
Topics	Deterministic clustering	$0	—
Competitor generation	1 Gemini call; Wikidata/Wikipedia APIs (free, throttled); ≤4 page fetches per candidate	$0	LLM tokens; a licensed web-search API if the disabled search source is enabled
Prompt generation	4 Gemini calls per set (intent model + 3 sections), plus retries; each Regenerate = 1 call; ~275–320 drafts validated locally	$0 (free tier)	LLM tokens per set or regeneration (a few cents at paid rates)
Observation: automatic	Gemini lane: 30 calls (a few extra early grounding attempts) on the free tier. ChatGPT lane: 30 `gpt-5-nano` calls, ~150 in / 1,200–2,200 out tokens including hidden reasoning. Groq/Cerebras: free but capped.	≈ $0.0005–0.001 per answer → **≈2–4¢ per 30-question ChatGPT run** (prepaid credit, $5 minimum)	Answers = prompts × engines × samples × frequency × markets. Web search adds ≈$0.01 per search plus page tokens (≈1–1.5¢ per answer). Bigger models cost 5–30× more. Paid Gemini rates were not assessed in this build.
Observation: by hand	Operator time in ChatGPT / Gemini apps	$0 marginal; 45–120+ min per 60 answers (plan estimate)	Analyst labour, or licensed collection partners with written rights
Analysis and measurement	Deterministic TypeScript: ≈16 ms per answer to analyse, ≈21 ms for the full metric computation (measured in audit)	$0	CPU only; no LLM judge by design
Storage, hosting, ops	Laptop, `data/sun.db`, `evidence-store/`	$0	Managed Postgres, object storage, workers and queue, SSE/realtime, auth, backups, security, legal/contract costs

Worked examples (estimates): 1 brand, 30 prompts × ChatGPT API × 5 samples × weekly ≈ 650 answers/month ≈ **$0.35–0.65** on gpt-5-nano without search. **Agency scale**, 100 brands × 100 prompts × 2 engines × 3 samples × weekly ≈ 260k answers/month ≈ **$130–260** at nano rates without search; ≈ **$2.6k–3.9k** with web search at ~1–1.5¢ per answer; plus hosting, and plus Gemini paid pricing if the free tier is outgrown. For a Gemini-key project, a full journey is ≈36–40 Gemini requests: 1 profile, 1 competitors, 4 prompts, ~30–33 observation.

13 · Architecture, stack, storage, versioning

Components and flow

```
Browser (React client pages)
⇅ JSON REST (Zod-validated) + SSE /api/projects/:id/events
Next.js 15 route handlers (single Node process)
→ domain modules: crawler → evidence → profile → landscape →
prompts → collection (runner) → analysis → measure
→ repository.ts per module (Drizzle) → SQLite WAL data/sun.db +
evidence-store/ files
→ outbound: target site (fetch / Playwright), Wikidata &
Wikipedia, Gemini, OpenAI, Groq, Cerebras
Consumer apps: opened only in the user's own browser tab
```

- **Background work** (crawl, landscape, prompt generation, collection) is fire-and-forget async code started by the API route, which returns 202. There is no Redis, queue or worker; a `jobs` table exists but is unused. Runner state lives in process memory, and orphaned rows are re-queued after a restart.
- **Events:** `appendEvent` inserts with an atomic per-project sequence, then pushes to SSE (15 s heartbeat, `Last-Event-ID` resume). The client also polls every 2.5 s and dedupes by sequence.
- **Repository seam:** the SQLite → Postgres swap is a dialect change (`text json` → `jsonb`).
- **Security:** SSRF guard resolves DNS and blocks private, loopback, link-local (incl. 169.254.169.254), CGNAT, multicast and IPv4-mapped IPv6, re-checked on every redirect and render. Zod at every boundary. Domain errors map to 400/409; anything else is a redacted 500. Keys stay server-side. No auth: local single operator.

Technology actually used

App	Next.js 15.5 App Router, React 19.1, TypeScript 5.9 (strict), Tailwind CSS 4.1, CVA / clsx / tailwind-merge
Data	SQLite via better-sqlite3 11 (WAL), Drizzle ORM 0.44 + drizzle-kit, 6 SQL migrations auto-applied; route policies seeded by `npm run dev`
Parsing	cheerio (HTML), tldts (domains), Zod (schemas), custom RFC 9309 robots, sitemap, JSON-LD / microdata / RDFa parsers
AI / APIs	`@google/genai` 1.9 (Gemini); OpenAI, Groq, Cerebras via plain `fetch`; Wikidata SPARQL and Wikipedia API
Browser	Playwright 1.55 Chromium: crawl fallback + E2E tests
Quality	Vitest 3.2 (44 test files: unit, integration, SSRF, benchmarks; 527 tests passing at last full run), Playwright E2E desktop Chrome + iPhone 13, ESLint 9, `npm run verify`, `npm run clean-room` (boots without keys)
Not used	No LLM judge, embeddings, vector DB, queue, Docker or cloud; `recharts` installed but unused (bars are CSS)

What is stored (27 tables + files)

Crawl	`projects, crawl_runs, crawl_pages, evidence_items, evidence_chunks` + page text files
Profile	`business_profile_versions` (profile JSON, provenance, user-edited paths, synthesis mode), `locations`
Landscape	`topics, entities, aliases, competitor_discovery_runs, competitive_set_versions`
Prompts	`prompt_context_snapshots, prompt_sets, prompt_candidates` (every draft, scores, rejections)
Collection	`collection_runs, prompt_runs, provider_responses, manual_capture_events` + raw answer files
Analysis	`observation_analyses` (full JSON with spans); `mentions, citations` (older parser); `metric_snapshots` unused (metrics computed live)
Platform	`provider_credentials, route_policies, events, jobs`

What is versioned or frozen

Object	Rule
Brand profile	Edits create version N+1 (partial versions update in place while crawling); confirmation recorded
Competitive set	Frozen snapshot + SHA-256 on **Confirm & continue**; identical content reuses the version
Prompt context	Snapshot + hash per generation
Prompt set	Version per generation; approval → `frozenHash`, immutable. Candidates keep every state; edits keep original text.
Answer	Raw text immutable + SHA-256; adapter, parser and policy versions
Analysis / metrics	Additive rows per `analysis_version` (current wins); `measure-policy@1` definitions returned with the numbers
Comparability	Two runs are comparable only if prompt set, competitive set, analysis version, metric policy, routes and sampling all match (function exists; no trend UI)

API surface (main)

```
POST /api/projects · POST /projects/:id/crawl · GET|PATCH
/projects/:id/profile · POST /profile/manual · GET
/projects/:id/evidence · GET /api/industries · POST
/landscape/generate · GET|PATCH /landscape · POST
/prompts/generate[?force=1] · GET /prompts · PATCH /api/prompt-
sets/:id · POST /prompt-sets/:id/approve · GET|POST
/providers/credentials · GET|POST /projects/:id/collection-runs
· GET /collection-runs/:id · POST /collection-runs/:id/routes ·
GET|POST /collection-runs/:id/collect · POST /prompt-
runs/:id/manual-capture · POST /prompt-runs/:id/retry|outcome ·
GET /projects/:id/measure · GET /provider-responses/:id · GET
/projects/:id/events
```

14 · Design decisions and build changes

Decision	Why	Change or evidence during the build
Evidence-first profile; confidence computed; model may only refine and must cite evidence ids	Stops invented facts; every field has a **Why?**; works without any key	Replaced "homepage text → LLM". Offerings stopped being nav labels, promo rails and customer logos.
Every profile field editable; edits versioned and protected from re-analysis	The business knows itself best; keeps inferred and corrected values auditable	Closed industry taxonomy and 12 archetypes so labels stay stable for downstream policy
Topics and competitors are suggestions the user confirms, rejects or adds; marketplaces never compared	Model and knowledge output is a hypothesis; the confirmed set *is* the share-of-voice denominator	Wikipedia co-membership (Armani, Dickies for Nike) → IDF category signature (Adidas, New Balance, ASICS). Adjacent rivals now frozen too (Adidas and Qdoba had been dropped).
Freeze context before prompts (competitive set + prompt context snapshots, hashed)	Regeneration can't silently use changed context; reproducible instrument	Added in the competitor and prompt intelligence rebuild
Exactly 30 prompts in three sections; split changed 21/5/4 → **10/10/10**	User brief: exactly three sections (Discovery, Trade Offs, Sentiment Analysis) shown equally; 30 is manually operable yet covers 4–6 topics	Policy versioned `prompt-selection@3-10-10-10`; old approved sets keep their split
Discovery is brand-blind (target *and* rivals banned)	A brand named in the question is not evidence the assistant volunteered it	Rival short names added to the ban list (e.g. "Pancheros")

Decision	Why	Change or evidence during the build
Customer intent model + one detail per prompt; templates only fill gaps	Round-2 audit: prompts read as templates (“Recommend the best apparel brands”)	Department words 8/10 → 0/10; prompts with a situation, need or budget 0/10 → 7/10; restaurant Discovery asks for a place, not a dish
Validation, scoring and selection are deterministic code; a model can't approve or bypass gates	Defensible, testable instrument; same inputs → same selection	Validators widened to accept real phrasing (“Torn between…”, “Looking for…”)
Collect merged into Measure; background lanes; Discovery first; pacing from provider limits; pause on quota	Old flow: one blocking request, fixed sleeps, a 6–7 min spinner, then 429s	First provisional figure ≈8–10 s; Gemini lane ≈2 min; quota exhaustion pauses with a resume time
Missing answers excluded, never counted as absence; **so far** labels; x-of-n everywhere; N/A ≠ 0%	A coverage gap must not become a competitive finding	Truncated answers flagged; Groq cap raised 1,200 → 3,000 tokens after half the answers were being cut off
Sections never blended; published metric definitions travel with the numbers	Vendors define visibility inconsistently (two common denominators diverge 2.18× on the same answers)	`measure/policy.ts`; the **?** drawer shows definitions verbatim
Rank only from explicit order; “no opinion” never positive; negation-aware clauses	Don't manufacture findings from sentence order or neutral facts	Negated mentions no longer forced review (9 → 0 of 30 Gemini answers)
Analysis versioned and additive over immutable raw text	Fix the parser without re-collecting	@1 → @9; same 60 Nike answers: analysable 51 → 60, share of voice 39% → 25% after short names and rivals were counted
By-hand route restored, with capture-state assertions	The only lawful way to observe the apps; “not captured” must not read as “no sources”	The route had been removed and left no key-free path; restored with an integration test
No consumer automation; UI names only ChatGPT and Gemini	Terms and robots rules; the marketing audience cares which assistant, not which API	Method details moved to the collapsed notes
ChatGPT lane: Groq gpt-oss (free) → Cerebras (402) → OpenAI `gpt-5-nano` (paid)	Free quotas too small or unreliable for a live demo; nano ≈0.1¢/answer	60/60 answers in ≈2 min 11 s observed
Gemini `flash-lite-latest` alias	Pinned models were retired for new keys; Flash's free quota was too small	Resolved model name recorded per answer
Local-first (SQLite, files, in-process jobs); repository pattern	Zero infrastructure for one operator; Postgres swap kept simple	Watcher excludes `data/` and `evidence-store/` (they had caused a reload loop)
No hardcoded brands, competitors, prompts or measurement values	Must generalise to any business	13-archetype benchmark; test lists live only in scripts

15 · Limitations and honest answers

Area	Limitation (as implemented today)	How to answer
Market & language	The start form's **Market** is stored but unused downstream. Language is recorded, but generation instructions are English; multilingual prompts are not validated.	“Markets come from site evidence and your edits; market targeting is next.”
Location	Target-market edits reach the prompt context snapshot, but generation prefers crawl-derived `locations` when present, so a market added by hand may not appear in prompts. No control or recording of where answers are asked from.	“Geography is prompt metadata, not execution location. Geo-controlled runs are production work.”
Changing context later	Re-confirming Landscape after prompts exist creates a new frozen version, but the Prompts page keeps the existing set (regenerating is API-only, `force=1`). Measure reads the project's *latest* frozen set, and linking uses currently confirmed entities, so re-confirming after collection changes the tracked set for that run.	“Add rivals, then regenerate and collect a new run; lineage shows versions. Old answers can be re-analysed.”
Sampling & fidelity	One sample per prompt (directional). Automatic lanes are APIs without web search, so no sources and citation metrics are mostly N/A. `gpt-5-nano` is not the ChatGPT app model.	“Directional baseline; ask by hand for app-true answers; search-enabled API is a paid switch.”
Manual capture	Relies on operator honesty. A saved answer can't be corrected in the UI (`succeeded` is terminal); one mis-pasted answer was seen in testing.	“Audit row per paste; production would add QA double-capture.”
Competitors & topics	Discovery can miss obvious rivals (e.g. Puma, Hoka at times) and include noise (On typed as marketplace). AI-only names may be unverified. Topics can be product lines (“Jordan”) or merchandising. Competitor verification ignores robots.txt.	“That's why they're suggestions; confirm, reject, add.”
Analysis	Lexical entity matching (no fuzzy or LLM tier); product names link only if stored as products. Sentiment is portrayal, not a fact-check. Lexical, not embedding, dedupe. Untracked list still has some noise.	“Deterministic and auditable by choice; ambiguous cases go to review, not into numbers.”
Prompts	Wording varies between regenerations (temperature 0.6–0.9). Regenerating a model-written prompt needs a Gemini key. Without a key, prompts are template-like.	“Each approved set is frozen; compare trends within one set.”
Platform	Single process: no queue, auth, multi-tenant, scheduling or trend view. Retention TTL not enforced. Keys pasted in the UI stored in plaintext locally. Grounding not gated (§11). Capability-gate flags on the project row are stale (e.g. `openaiApiEnabled=false`). Landing page shows illustrative “Position/Coverage” cards that Measure doesn't show as cards.	“Phase 2 is a local measurement build; these are the productionisation items.”
Demo ops	Dev mode compiles routes on first hit (seconds of delay). Groq and Cerebras free tiers can pause.	Use `npm run build && npm start`; OpenAI + Gemini keys set.

16 · Responsive and mobile behaviour

Tailwind breakpoints: **sm** 640 px, **md** 768, **lg** 1024. Every screen is fluid; nothing needs horizontal scrolling except the stage tabs.

- **Landing:** section links show from 1024 px. Below 640 px they, and **Start free analysis**, move into a menu button. Between 640 and 1023 px only the Start button shows, with no section links (gap). Hero,

17 · Quick answers to likely questions

Real or mocked?	Real crawls and real provider answers, stored locally. Only landing previews are illustrative. No brands, competitors or prompts are hardcoded.
Are you scraping ChatGPT?	No. Automatic = official APIs; the apps are only asked by a person who pastes the answer back.

sections and cards stack. The website form stacks the input above the button below 640 px.

- **Journey nav:** brand and stage tabs stack; the tabs scroll sideways.
- **Understand:** single narrow column; chips and buttons wrap.
- **Landscape:** row actions drop under the text; the add-competitor row (name / relationship / button) stacks on phones; the **Why?** panel goes from 3 columns to 1.
- **Prompts:** filters stack; row buttons wrap under the prompt; **Why?** grid 1 → 2 → 4 columns; the three section counters stay in one row.
- **Measure:** headline cards 1 → 2 (sm) → 4 (lg). Section panels stack below 1024 px (5/7 split above). The All/ChatGPT/Gemini switch wraps under the title. The provisional banner stays sticky. Drawers are full-width on phones. Console lanes 1 → 2 columns; the paste form's 3 selects stack.
- Honours reduced-motion. Keyboard focus rings; drawers close with Esc.
- **Tested:** Playwright runs the journey tests on Desktop Chrome and an iPhone 13 profile, plus an explicit 1280×800 and 390×844 check of the start form. Measure on mobile has no automated test, so check it visually before a demo.

Is “ChatGPT, automatic” the app?	No: OpenAI `gpt-5-nano` via API, stated in the method notes. Ask by hand for app-true answers.
Why 30, and 10/10/10?	Three constructs of equal weight; covers 4–6 topics; still operable by hand; comparable to commercial starter sets. Directional with 1 sample.
Why brand-neutral Discovery?	Naming the brand makes a mention likely; only unprompted recommendation measures visibility.
Share of voice?	Brand presence events ÷ presence events of the brand + frozen rivals, in Discovery answers, once per answer.
Failed answers?	Excluded from both sides of every fraction and shown as coverage; lanes pause and resume.
No sources shown?	Automatic lanes don't search. Paste sources by hand; web-search API is a paid option; Gemini grounding is restricted.
Wrong competitor suggested?	Reject it and add yours. Only confirmed rivals are tracked; the set is frozen per version.
Can I edit prompts?	Yes; every edit is re-validated (you can't leak the brand into Discovery). Approval freezes the set.
LLM judging answers?	No: a deterministic, versioned, negation-aware parser; re-analysable without re-collecting.
Hallucinated profile?	Fields must cite evidence ids; confidence is computed; everything is editable.
Site blocks crawling?	robots.txt respected; the page says why; the profile can be completed by hand.
Time?	Crawl ~10–40 s · landscape ~1–1.5 min · prompts ~10–14 s · 60 answers ~2 min · first figure ~8–10 s.
Cost?	≈2–4¢ per ChatGPT API run; everything else free tier or local (§12).
Scale-up plan?	Postgres, queue and workers, per-route token buckets, object storage, scheduled multi-sample runs, licensed or authorised collection, auth and tenancy.